

In the adiabatic limit, i.e. when the frequency of the potential modulation is small compared to the characteristic times for traversal and reflection of electrons, the photon side band $E_F \approx E_r \pm \hbar\omega$ broadens the current peak instead of forming new peaks as in Ref.5. In principle, without the external modulations the resonant tunnelling current is broadened due to spin mixing at the DWs. If the spin-mixing amplitude is finite (i.e., $\Delta \neq 0$), spin-up carriers transverse resonantly the DWs and the conductance peaks are broadened by the spin-mixing mechanism. For very narrow DWs the residual broadening can then be attributed to the side-band formation.

In the theory of an adiabatic quantum pump, the nearest sidebands corresponding to particles which have gained or lost a modulation quantum $\hbar\omega$ contribute to the pumped current. Hence, the pumped current peak is broadened by $2\hbar\omega$ at the resonant levels. The width of peaks in the linear-response conductance due to spin mixing at the DWs is enhanced for wider DWs (cf. Fig. 5). In comparison, the broadening of the pumped current peaks caused by the sideband formation is less pronounced for varying DW extensions. For very sharp DWs however the effect of the side-band formations might be as large as the contribution from the DWs spin mixing.

To demonstrate the quantum pump noise properties in this particular structure, we present the numerically obtained pump current I , shot noise S , heat flow H , and relative pump noise F as functions of the Fermi energy in Fig. 3. The properties for different DW distance L are compared. The pump current, shot noise and heat flow show peak structure when the Fermi energy matches with the resonant energy levels of the structure. The peak position changes accordingly as the resonant levels shift to lower energy when we increase the DW distance L . The ratio between the pump shot noise and the heat flow F reflects the strength of the quantum correlation of the system. F shows bifurcated peaks with the intermediate minima at the resonant Fermi energies.

The pump noise properties can be interpreted as follows. At the resonant Fermi energy, transport processes and thus the electron-electron correlation achieve maximal strength. The nonequilibrium quasi-electrons and holes created by the oscillating scatterer move in different directions to generate a net dc current. Therefore, the antibunching correlation between electrons and holes always exerts a negative contribution to the shot noise regardless of the direction of the dc current, which gives rise to the minimum valley in the relative noise. At the edges of the resonant state, there is a slight heat flow in the dissipation regime